ANNA UNIVERSITY, CHENNAI-600 025

COLLEGE OF ENGINEERING GUINDY

A CNN BASED DEEP LEARNING MODEL FOR DETECTING P2P BOTNETS USING FLOW FEATURES

A PROJECT REPORT

Submitted by

A T PRANAV VARSHAN 2025165034

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ABSTRACT

Botnets are one of the major cybersecurity threats in modern networks, especially with the rapid growth of Internet-connected devices. Among different botnet architectures, Peer-to-Peer (P2P) botnets are particularly difficult to detect because they do not rely on a centralized command server and instead communicate directly with other infected devices. This decentralized structure makes traditional detection methods less effective.

This project presents a deep learning-based approach for detecting P2P botnet activities using network traffic flow features. The model is developed using a Convolutional Neural Network (CNN), which is capable of automatically learning patterns and relationships within network traffic data. The BoT-IoT dataset is used for training and evaluation, and a subset of important network flow features is selected to represent the traffic behavior.

In the proposed system, the dataset is first preprocessed by cleaning the data, normalizing the feature values, and selecting relevant flow features. The processed data is then divided into training and testing sets. The CNN model is designed with convolution, activation, pooling, and fully connected layers to extract meaningful patterns from the network traffic flows. The final classification layer predicts whether the traffic corresponds to normal activity or botnet attack behavior.

The experimental results demonstrate that the CNN-based model can effectively detect botnet traffic with high accuracy and reliability. The approach shows strong potential for improving network security by enabling automated detection of malicious activities in large-scale network environments.

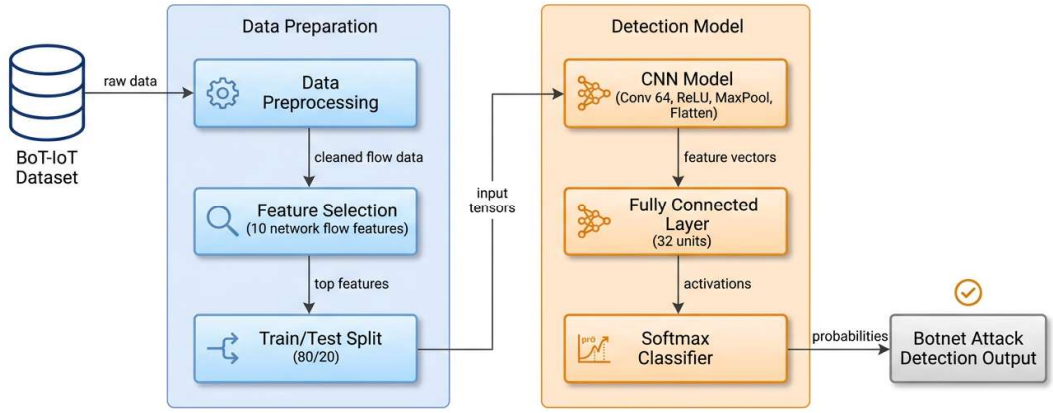
OBJECTIVES OF THE PROJECT

- To analyze network traffic and identify malicious botnet activities.
- To extract important network flow features that help in detecting botnet behavior.
- To develop a Convolutional Neural Network (CNN) model for identifying P2P botnet attacks.
- To classify network traffic into normal traffic and botnet traffic.
- To evaluate the performance of the proposed model using accuracy, precision, recall, and F1-score.

PROBLEM STATEMENT

Botnets are one of the major cybersecurity threats where compromised devices are controlled by attackers to perform malicious activities such as Distributed Denial of Service (DDoS), data theft, and network attacks. Traditional detection methods such as signature-based and rule-based approaches often fail to detect modern Peer-to-Peer botnets because these botnets use decentralized communication structures and encrypted traffic. This makes it difficult to identify malicious activities using conventional techniques. Therefore, there is a need for an intelligent detection mechanism that can analyze network traffic patterns and identify botnet behavior efficiently. The proposed project addresses this issue by using a CNN-based deep learning model to detect P2P botnet attacks using network flow features.

OVERALL ARCHITECTURE



1. Data Preparation

This phase handles the ingestion and refinement of raw network traffic data so the neural network can efficiently process it.

- Data Preprocessing: The raw data from the BoT-IoT dataset is cleaned to handle missing values, normalize data scales, and remove noise, resulting in standardized flow data.
- Feature Selection: To optimize the model's performance and reduce computational overhead, the pipeline isolates the top 10 most critical network flow features rather than processing every single packet attribute.
- Train/Test Split (80/20): The refined dataset is divided into two sets. 80% of the data is used to train the model, while the remaining, unseen 20% is held back to test its accuracy. The data is then formatted into input tensors.

2. Detection Model

This phase acts as the classification engine of the system, taking the prepared tensors and evaluating them for malicious activity.

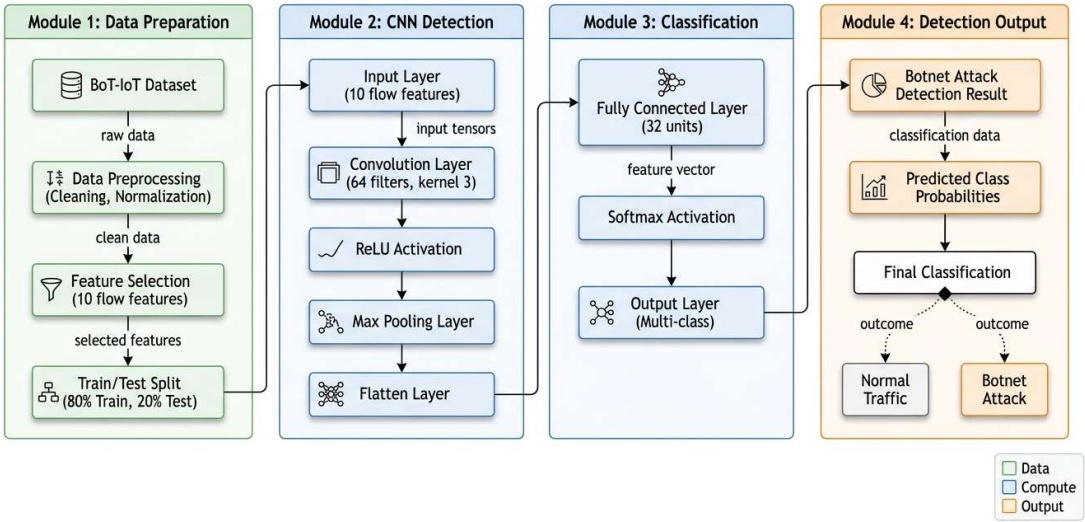
- CNN Model: A Convolutional Neural Network extracts complex patterns from the network traffic. It utilizes a 64-filter Convolutional layer (Conv 64) for feature extraction, a ReLU activation function to introduce non-linearity, a MaxPool layer to downsample the spatial dimensions, and a Flatten layer to convert the multi-dimensional output into a 1D array (feature vectors).
- Fully Connected Layer (Dense Layer): Consisting of 32 units, this layer takes the extracted feature vectors and learns the complex, non-linear combinations of these features to determine attack signatures, outputting activations.
- Softmax Classifier: As the final layer, the Softmax function calculates the mathematical probabilities for the different classes (e.g., normal traffic vs. specific botnet vectors like DDoS or keylogging).

Final Output:

Based on the highest probability generated by the Softmax classifier, the system delivers the final Botnet Attack Detection Output.

DETAILED DESIGN (Module Wise)

The proposed system for P2P botnet detection is divided into four main modules. Each module performs a specific function in processing the network traffic data and detecting malicious activities.



Module 1: Data Preparation:

This module is responsible for collecting and preparing the dataset before feeding it into the deep learning model.

BoT-IoT Dataset

The system uses the BoT-IoT dataset, which contains network traffic data collected from IoT environments. The dataset includes both normal traffic and botnet attack traffic. Each network flow record contains several attributes describing the behavior of the connection.

The dataset serves as the raw input data for the system.

Data Preprocessing

The raw dataset often contains noise, missing values, and inconsistent data. Therefore, preprocessing is performed to clean the dataset and make it suitable for machine learning.

The preprocessing steps include:

- Removing duplicate records
- Handling missing values
- Normalizing feature values
- Converting categorical features into numerical values

After preprocessing, the dataset becomes clean data, which improves the accuracy of the detection model.

Feature Selection:

From the large number of available features in the dataset, only the most relevant features are selected for the detection process.

In this system, 10 network flow features are selected, such as:

- Standard deviation of flow rate (SD)
- Number of incoming flows (nIn)
- Number of outgoing flows (nOut)
- Minimum duration of connection (Min_dur)
- Maximum duration of connection (Max_dur)
- Flow state (State)
- Mean duration (Mean)
- Outgoing packet rate (OutRate)
- Incoming packet rate (InRate)
- Argus sequence number (Seq)

Selecting important features helps in reducing computational complexity and improving model performance.

Train/Test Split:

The prepared dataset is divided into two parts:

- 80% Training Data – used to train the CNN model
- 20% Testing Data – used to evaluate the performance of the trained model

This ensures that the model can be tested on unseen data to measure its accuracy.

Module 2: CNN Detection:

This module contains the Convolutional Neural Network (CNN) used to detect botnet attacks from network traffic features.

Input Layer

The selected 10 network flow features are provided to the CNN model as input. The features are reshaped into a format suitable for the CNN, usually represented as a 1D feature vector.

Convolution Layer

The convolution layer applies 64 filters with kernel size 3 to the input data.

This layer performs a convolution operation that scans through the feature vector and extracts important patterns related to botnet behavior.

It helps the model learn hidden relationships between different network flow features.

ReLU Activation

After convolution, the Rectified Linear Unit (ReLU) activation function is applied.

ReLU introduces non-linearity into the model and converts negative values to zero. This improves the learning capability of the neural network.

Max Pooling Layer

The max pooling layer reduces the dimensionality of the feature maps generated by the convolution layer.

It selects the maximum value from each region of the feature map, which helps in:

- Reducing computational cost
- Removing unnecessary information
- Retaining important features

Flatten Layer

The flatten layer converts the multi-dimensional feature maps into a one-dimensional vector.

This vector representation is then passed to the classification module.

Module 3: Classification:

This module is responsible for classifying the network traffic based on the features extracted by the CNN.

Fully Connected Layer

The flattened feature vector is passed to a fully connected neural network layer with 32 neurons.

This layer combines the extracted features and learns the complex patterns that distinguish botnet traffic from normal traffic.

Softmax Activation

The softmax function converts the output of the neural network into probability values for each class.

For example:

Normal traffic → 0.20
Botnet attack → 0.80

These probabilities represent the likelihood of the traffic belonging to each class.

Output Layer

The output layer performs multi-class classification based on the probabilities generated by the softmax function.

The model predicts whether the input network flow belongs to normal traffic or botnet attack traffic.

Module 4: Detection Output:

This module produces the final detection result.

Botnet Attack Detection Result

The classification result from the previous module is used to determine the final output.

The system provides:

- Predicted class probabilities
- Final classification result

Final Classification

Based on the predicted probabilities, the system decides whether the traffic belongs to:

- Normal Traffic
- Botnet Attack

This information can be used by network administrators to identify infected devices and prevent cyber attacks.